ERIN RACHEL THOMAS

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M.Tech Cyber Security and Digital Forensics

Education

Vellore Institute of Technology 2025 – 2027
Master of Technology in Cyber Security & Digital Forensics *CGPA: 8.12*

Muthoot Institute of Technology and Science 2021 – 2025
Bachelor of Technology in Computer Science and Engineering *CGPA: 6.95*

Projects

AI in cyber warfare *Cybersecurity, Machine Learning, Python*
September 2025 - April 2026

- Conducting research on the role of artificial intelligence in modern cyber warfare and threat prediction.
- Studying machine learning approaches for cyber attack detection and classification.
- Analyzing predictive techniques for identifying threats targeting critical digital infrastructure.
- Exploring the impact of AI-driven cybersecurity strategies in political and national security contexts.

Machine learning-based Intrusion Detection System *Python, FFNN architecture, TensorFlow, Pandas*
August 2025 - November 2025

- Developed and implemented IDS using deep learning architectures.
- Designed the system to effectively detect DoS and DDoS attacks using CSE-CIC-IDS2018 Dataset.
- Removal of irrelevant features to enhance model training efficiency and accuracy.
- Reducing false negatives and ensuring robust network integrity against evolving cyber threats.

Online exam proctoring system *HTML, CSS, YOLO, Haar Cascade, LBPH, Python, Sql*
July 2024 - April 2025

- Implemented YOLO for object detection such as phone.
- Uses Haar Cascade and LBPH algorithm for face recognition.
- Real time image and video processing to detect malpractice
- Documented the number of head movements and malpractice count

Smart traffic management system *YOLO, Python, OpenCv*
February 2024 - June 2024

- Analysis of traffic signals based on the maximum number of cars at a signal.
- Tracking the traffic signal using video fragmentation and image processing.
- Implemented real-time frame processing, bounding box visualization and vehicle counting optimizing performance for traffic analysis applications.
- A YOLO-based object detection system to identify and track vehicles in video feeds.

Skills

Languages: Python, SQL, HTML, CSS
Tools & OS: Kali Linux, FTK Imager, Wireshark, Nmap, Burp Suite, Microsoft
Security Domains: Digital Forensics, Network Security, AI Security, Vulnerability assessment
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Object Oriented Programming, Database Management System, Computer Networking.
Soft Skills: Problem Solving, Critical Thinking, Adaptability, Technical Writing, Presentation, Speaking

Certifications

Cyber security course, 2024 – Concepts about Encryption and Decryption, basics on Cyber Security
Crash course on Python Coursera, 2023 – Foundational programming concepts, OOPs concepts
Business English in Cambridge, 2022 – Merit

Extra-Curricular

Part of organizing committee for tech fest